MATH 128a: Homework #12

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Due by: **Wednesday, December 3, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Numerical Quadrature 1

Derive a formula for approximating

$$\int_1^3 f(x)dx,$$

in terms of $f(0)$, $f(1)$ and $f(4)$. It should be exact for all polynomials of degree up to 2.

Problem 2: Numerical Quadrature 2

Derive the Newton-Cotes formula for

$$\int_0^1 f(x)dx,$$

based on the Lagrange interpolation polynomial at the nodes -2, -1 and 0. Apply this result to evaluate the integral when $f(x) = \sin(x)$.

Problem 3: Numerical Quadrature 3

Find a quadrature rule

$$\widehat{I}_{[0,1]}(f) = w_0 f(0) + w_1 f(1),$$

that approximates the integral

$$I_{[0,1]}(f) = \int_0^1 f(x)dx,$$

and that is exact for all functions of the form $f(x) = ae^x + b\cos(\pi x/2)$; $a, b \in \mathbb{R}$.

Problem 4: Numerical Quadrature 4

a) Use the Lagrange interpolation polynomial to derive a quadrature rule of the form

$$\widehat{I}_{[0,1]}(f) = w_0 f(1/3) + w_1 f(2/3),$$

that approximates the integral

$$I_{[0,1]}(f) = \int_0^1 f(x)dx.$$

b) Transform the resulting formula to one for integration over $[a, b]$, $a < b$.

c) Apply this result to evaluate

$$\int_0^{\pi} \sin(x)dx.$$

Compare with the exact value of the integral.

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